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Transfinite Iteration of Operator Transforms and Spectral Projections in Hilbert and Banach Spaces

Faruk Alpay* Taylan Alpay† Hamdi Alakkad‡

Abstract

We develop an analytic framework for ordinal-indexed, multi-layer iterations of bounded linear operator transforms on Hilbert and reflexive Banach spaces. Given a normal operator A on a Hilbert space with a family of K polynomial or holomorphic transforms $\Phi_k(T) = p_k(T)$ satisfying $\sup_{\lambda\in\sigma(A)} |p_k(\lambda)| \leq 1$, $p_k(1) = 1$, and with the *peripheral spectrum* fixed under p_k, we prove that the transfinite iteration $A^{(\alpha)}$ (applying $\Phi_1, \ldots, \Phi_K$ in cycles) converges in the strong operator topology by some countable stage $\alpha < \omega_1$ to an idempotent P that commutes with A. The limit P is the spectral projection of A onto the joint fixed-point set and $\sigma(P) = \{0, 1\}$. We also establish a precise spectral mapping property for every stage of the iteration under the functional calculus. In a reflexive Banach space setting, for a power-bounded Ritt operator (or sectorial generator of a bounded analytic semigroup) A admitting a bounded H^∞ functional calculus, an analogous multi-layer iteration $\Phi_K \circ \cdots \circ \Phi_1$ (with $\sup |\varphi_k| \leq 1$, $\varphi_k(1) = 1$) converges in strong operator topology to a bounded projection P commuting with A. The range of P consists of vectors invariant under all transforms (in particular, $AP = PA$). We provide proofs via the spectral theorem (normal case) and via resolvent and ergodic techniques (Ritt/sectorial case), along with explicit examples and counterexamples. A Fejér-monotonicity argument yields an ordinal bound $\leq \omega$ (countable) for stabilization. We construct cases achieving the extremal countable stage and show that allowing non-commuting layers or dropping the Ritt/sectorial assumptions can destroy convergence. Connections with classical mean ergodic projections [1, 2] and stability theorems of Arendt–Batty and Lyubich–Vũ [2, 7, 8] are discussed.

1 Introduction

Iterative methods for projecting onto invariant subspaces and spectral subspaces are a classical theme in operator theory and ergodic theory [1, 2]. A basic instance is the power method: if A is a contraction on a Hilbert space, then A^n often converges (strongly) to a projection onto the fixed-point space $\ker(I - A)$ by the mean ergodic theorem. More generally, for a power-bounded operator with spectral radius 1, classical results (e.g. the Katznelson–Tzafriri theorem and the Arendt–Batty–Lyubich–Vũ stability theorem) give conditions under which $A^n \to P$ as $n \to \infty$, where P is the projection onto the eigenspace at eigenvalue 1. In continuous time, similar phenomena occur: a C_0-semigroup $T(t) = e^{tA}$ may converge strongly to a projection as $t \to \infty$ under spectral assumptions on the generator A (e.g. no spectrum on the imaginary axis except 0). Such limits are *ergodic projections* in the sense of mean ergodic theory.

In this work, we introduce a broad iteration scheme that generalizes these one-parameter ergodic results to *multi-layer, transfinite iterations of operator transforms*. Concretely, we consider

*Department of Mathematics, Lightcap Institute. alpay@lightcap.ai
†Department of Aerospace Engineering, Turkish Aeronautical Association.
‡Department of Engineering, Bahçeşehir University.

a family of K operator functions $\Phi_1, \ldots, \Phi_K$ ("layers") and define an ordinal-indexed sequence of operators: $A^{(\alpha+1)} = \Phi_K(\cdots \Phi_1(A^{(\alpha)}) \cdots)$ for successor ordinals, and at a limit ordinal λ we define $A^{(\lambda)} = \text{SOT-}\lim_{\alpha\to\lambda} A^{(\alpha)}$ assuming the limit exists in the strong operator topology. This transfinite iterative process continues (if necessary) through countable ordinals. We show that under suitable analytic conditions on the Φ_k, the sequence $(A^{(\alpha)})_{\alpha<\omega_1}$ stabilizes by some countable stage: there is an ordinal $\alpha_* < \omega_1$ such that $A^{(\alpha)} = A^{(\alpha_*)}$ for all $\alpha \geq \alpha_*$. In fact, we obtain explicit bounds on α_* in terms of the properties of the transforms (often $\alpha_* \leq \omega$, the first countable ordinal). The limiting operator $A^{(\alpha_*)}$ is a bounded idempotent (projection) P that commutes with the original operator A [7, 8].

We treat two primary frameworks:

(I) Hilbert space (normal A): A is a normal operator on a complex Hilbert space H. We assume $\Phi_k(T) = p_k(T)$ for $k = 1, \ldots, K$, where each p_k is a polynomial or a bounded holomorphic function on an open set containing $\sigma(A)$. These p_k are **Schur holomorphic functions** on the spectrum, meaning $\sup_{\lambda\in\sigma(A)} |p_k(\lambda)| \leq 1$, and they fix the peripheral spectrum: in particular $p_k(1) = 1$, and for every $\lambda \in \sigma(A)$ with $|\lambda| = 1$, if $|p_k(\lambda)| = 1$ then $p_k(\lambda) = \lambda$. (This is the *peripheral fixed-point property.*) Such Φ_k clearly commute with each other (since they are functions of A or its iterates), so the multi-layer composition $\Phi_K \circ \cdots \circ \Phi_1$ applied to $A^{(\alpha)}$ is unambiguous.

(II) Banach space (power-bounded A): A is a power-bounded linear operator on a reflexive Banach space X (discrete-time case), or $-A$ generates a bounded analytic C_0-semigroup (continuous-time case). We assume A admits a bounded H^∞ functional calculus on the spectral domain (a Stolz disk containing $\sigma(A)$ in the discrete case, or a sector containing $\sigma(A)$ in the continuous case [9, 10]). We take $\Phi_k(T) = \varphi_k(T)$ where each φ_k is a rational or holomorphic function on the domain of A (unit disk or sector) such that $\sup |\varphi_k| \leq 1$. These conditions ensure $\varphi_k(A)$ is well-defined (via the functional calculus) and $\|\varphi_k(A)\| \leq C$ for some constant C, and again φ_k fixes the point 1 on the boundary of $\sigma(A)$ (the unit circle or spectral boundary). We consider the same transfinite iteration which is well-defined since each φ_k preserves the class of operators (Ritt or sectorial) and the $\varphi_k(A^{(\alpha)})$ all commute (being functional calculus images of a common original A or its iterates).

Under these assumptions, our main results can be summarized as follows.

2 Convergence results in the Hilbert space (normal) case

Standing assumptions (normal case). Let $A \in B(H)$ be a normal operator on a Hilbert space H, with spectral measure E_A and spectrum $\sigma(A) \subseteq U$ (where $U \subset \mathbb{C}$ is an open set). Let $p_1, \ldots, p_K$ be functions holomorphic on U (polynomials are a special case) with a uniform bound $\sup_{\lambda\in\sigma(A)} |p_k(\lambda)| \leq 1$. Define the composite function $f := p_K \circ \cdots \circ p_1$, and denote by f^m the m-fold composition ($f^m := f \circ f \circ \cdots \circ f$, m times). We apply these to A via the functional calculus: write $A^{(m)} = f^m(A)$ for $m \in \mathbb{N}$ (so $A^{(0)} = A$ and $A^{(m+1)} = f(A^{(m)}) = f^{m+1}(A)$). At the first limit stage $m \to \infty$, if the strong operator limit exists we denote $A^{(\omega)} = \lim_{m\to\infty} A^{(m)}$. Note that by the properties of the normal functional calculus, $A^{(m)} = (f^m)(A)$, and two standard properties will be used tacitly throughout:

- **(Composition in the calculus).** If h is a bounded Borel function on $\sigma(A)$ and g is a bounded Borel function on $h(\sigma(A))$, then $g(h(A)) = (g \circ h)(A)$.
- **(Dominated convergence in the calculus).** If g_n are bounded Borel functions on $\sigma(A)$ with $\sup_n \|g_n\|_\infty < \infty$ and $g_n(\lambda) \to g(\lambda)$ pointwise on $\sigma(A)$, then $g_n(A) \to g(A)$ in strong operator topology.

(Proofs of these facts can be found in Appendix A as Lemmas .6 and .7 respectively.)

We now present two main theorems (labeled A' and B') for the Hilbert space case, concerning convergence to a projection (Theorem A') and the spectral behavior along the iteration (Theorem B'). These results are formulated under general conditions that will later be verified by more specific assumptions (see Lemma 2.6 and Corollary 2.4 below).

Theorem 2.1 (A' – Convergence via function iteration $\implies$ spectral projection)**.** Suppose (under the standing setup above) that the sequence of functions f^m is uniformly bounded and converges pointwise on $\sigma(A)$ to a (Borel) limit function $\chi : \sigma(A) \to \mathbb{C}$. That is, assume: $\sup_m \|f^m\|_\infty < \infty$ and $\lim_{m\to\infty} f^m(\lambda) = \chi(\lambda)$ for all $\lambda \in \sigma(A)$. Then the strong limit $A^{(\omega)}$ exists and $A^{(\omega)} = \chi(A)$. If, moreover, χ takes only the values 0 and 1, then $P := \chi(A)$ is an orthogonal projection that commutes with A, and $\operatorname{Ran} P = E_A(\chi^{-1}(\{1\}))H$. (Indeed, $\sigma(P) = \{0, 1\}$ unless $P = 0$ or $P = I$.)

Proof. By the dominated convergence property of the normal calculus, $f^m(A) = (f^m)(A) \to \chi(A)$ strongly as $m \to \infty$. If χ takes only the values 0 and 1, then $\chi(A) = \int \chi \, dE_A = E_A(\chi^{-1}(\{1\}))$, which is an orthogonal projection (since $\chi^{-1}(\{1\})$ is a measurable subset of $\sigma(A)$) and clearly commutes with A (being a function of A). The range of $\chi(A)$ is $\operatorname{Ran} E_A(\chi^{-1}(\{1\}))$, which is exactly the subspace of H where A acts with spectrum in $\chi^{-1}(\{1\})$. This shows the stated description of P and its range. □

Theorem 2.2 (B' – Spectral mapping along the iteration)**.** Under the same standing assumptions, one has:

(i) (Finite stages). For each finite $m \geq 1$, $\sigma(A^{(m)}) = \sigma(f^m(A)) = f^m(\sigma(A))$. In other words, spectral mapping holds as an equality at every finite iterate (since f^m is holomorphic on a neighborhood of $\sigma(A)$).

(ii) (Limit stage). If $f^m \to \chi$ pointwise on $\sigma(A)$ with $\sup_m \|f^m\|_\infty < \infty$ (so that $A^{(\omega)} = \chi(A)$ exists by Theorem 2.1), then $\sigma(A^{(\omega)}) \subseteq \overline{\chi(\sigma(A))}$. In particular, if $\chi \in \{0, 1\}$ (so that $A^{(\omega)} = P$ is a projection), then $\sigma(A^{(\omega)}) \subseteq \{0, 1\}$ (and in fact $\sigma(P) = \{0, 1\}$ unless P is trivial 0 or I).

Proof. Part (i) is immediate: since f^m is holomorphic on an open set containing $\sigma(A)$, the spectral mapping theorem for normal operators gives $\sigma(f^m(A)) = f^m(\sigma(A))$ (no inclusion restriction is needed in the normal case).

For part (ii), we know $A^{(\omega)} = \chi(A)$ by Theorem 2.1. By functional calculus, $\sigma(\chi(A)) = \overline{\chi(\sigma(A))}$ if χ were continuous on $\sigma(A)$. However, χ might be just a bounded Borel function (for example, χ could be discontinuous if it jumps between 0 and 1 on different parts of the spectrum). In general, one only has $\sigma(\chi(A)) = \operatorname{ess\,range}_{E_A}(\chi)$, the essential range of χ with respect to the spectral measure E_A. Proposition 2.3 below provides a proof of this fact. Since the essential range $\operatorname{ess\,range}_{E_A}(\chi)$ is always a subset of the closure of the pointwise range $\overline{\chi(\sigma(A))}$, we have $\sigma(A^{(\omega)}) \subseteq \overline{\chi(\sigma(A))}$. Finally, if χ only takes values $0, 1$, then $\chi(\sigma(A)) \subseteq \{0, 1\}$ and hence $\sigma(P) \subseteq \{0, 1\}$. (In fact, as noted, $P = \chi(A)$ is a nonzero projection unless $A^{(\omega)}$ turned out trivial 0 or I, so $\sigma(P)$ must equal $\{0, 1\}$.) □

We isolate here the general fact used above, for completeness:

Proposition 2.3 (Essential-range description of $\sigma(\chi(A))$)**.** *Let A be a normal operator with spectral measure E_A, and let $\chi : \sigma(A) \to \mathbb{C}$ be a bounded Borel function. Then $\sigma(\chi(A)) = \{\lambda \in \mathbb{C} : E_A(\{\omega \in \sigma(A) : |\chi(\omega) - \lambda| < \varepsilon\}) \neq 0$ for all $\varepsilon > 0\}$.*